Primordial Outpost Reborn

Oxygen Not Included

Modlist & Guide

Build 744825 (2026-07-28) - 2026

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Installation Guide

Game Prep

This guide targets **Oxygen Not Included** running the base game plus the two expansions you already own: **Spaced Out!** and the **Bionic Booster Pack**. Install the game from Steam together with both DLCs before touching any mods.

1. On the Steam store page, install **Oxygen Not Included**, then make sure **Spaced Out!** and the **Bionic Booster Pack** are installed as well.
2. Launch the game once, all the way to the main menu, so Steam and the game finish their first-time setup.
3. On the main menu, confirm the build number reads **Build 744825**, released July 28, 2026. This is the build the entire guide is written against.

> **Note:** The **Aquatic Planet Pack** (launched with the June 11, 2026 update), the **Frosty Planet Pack**, and the **Prehistoric Planet Pack** are **not** part of this guide. You do not need them, and any mod that lists one of them as a requirement is incompatible with your setup.

> **Tip:** Build numbers change whenever the game updates. Every time Oxygen Not Included downloads an update, re-check the main menu and re-confirm that it still shows Build 744825 — a future patch may shift the mods you are using.

Mod Updater

The organizer this guide uses is the workshop mod-updater tool **Mod Updater** — Mod Updater on the Steam Workshop. It keeps your subscribed mods current and prevents Steam's Workshop cache from serving stale versions.

> **Note:** The tool's attribution is recorded in the project's STATUS.md.

To install it, just subscribe on the Workshop page. Oxygen Not Included downloads the mod on its next launch; there is nothing else to set up.

Once subscribed, Mod Updater lives inside the in-game Mods dialog:

1. It shows each mod's last-update date.
2. It flags mods that are out of date.
3. When at least one mod is outdated, an **Update All** button appears at the bottom of the Mods dialog.
4. When mods are out of date, the Main Menu shows a warning telling you so.

Enabling and disabling mods is unchanged: that stays in the in-game Mods menu. Mod Updater only handles updating.

> **Warning:** Mod Updater works with a **Steam-purchased** copy of Oxygen Not Included only. It does not work if you bought the game from the Epic Games Store or WeGame.

> **Warning:** On Windows, OneDrive can still break mod updates even with Mod Updater installed, causing a mod update/restart loop. The recommended mitigation is to turn OneDrive off for your Documents folder.

Fetching the Wave's Mods

The mods in this guide are added per wave, not all at once. Each wave's Modlist section holds that wave's mod cards, and each card carries the Workshop URL for its mod. In this release the cards are pending addition.

To fetch a wave's mods, open each card's Workshop page in turn and subscribe. Mod Updater then keeps every subscribed mod updated for you.

Version Compatibility

The Steam Workshop does not publish a per-mod build number, so there is no exact "works with Build 744825" stamp on any page. The best evidence of build compatibility is the mod's **last updated** date: a mod updated close to Build 744825 (July 28, 2026) is the safest bet, because it was most recently checked against the current game. A mod that has not been touched in months may still work, but it is riskier.

DLC requirements matter just as much as dates. A mod that requires a DLC you do not own is incompatible even though Steam will happily let you subscribe — see the Game Prep note for the DLCs outside this guide's scope.

When the game updates, re-check your mods. A new game build can break mods that were fine the day before, and Mod Updater flags exactly that.

> **Tip:** Read the mod's Workshop page yourself before subscribing. The last-update date shown on the page, together with the DLC tags, is the signal to trust — not titles or third-party "updated" badges.

Launch & Verify

1. Start the game.
2. Open the Mods dialog. It lists every installed and enabled mod — treat it as ground truth for what is active.
3. Confirm nothing is flagged as outdated. If a flag appears, use **Update All** and restart.
4. Start a **new** game for the wave you intend to play, then play the opening stretch and confirm the mods behave the way the wave guide describes.

> **Tip:** Do not trust a mod list from memory. The Mods dialog is the only authoritative view of what is actually enabled, and it takes one glance to confirm before each session.

Per-Wave Switching

Each wave is its own new game with its own mod subset. Switching waves is a menu chore, not a reinstall:

1. Open the Mods menu.
2. Enable the new wave's mods and disable the previous wave's content mods.
3. Start a new game for the wave.

Wave 0 carries the quality-of-life base that stays active throughout. Content mods stay scoped per wave and are only enabled when that wave's design calls for them.

> **Warning:** Checkbox errors break the wave design. If recipe or content mods from one wave are carried into a wave that should not have them, the intended progression is lost and the game plays differently than the guide describes. Review the Mods menu before starting each new game.

Tip: Because Mod Updater handles updates for every subscribed mod, wave switching never requires unsubscribing or re-downloading anything. The Mods menu checkboxes are the entire workflow.

Waves & New-Game Setups

Wave 0: The First Breathe

Wave 0: How to Play

Three rookie duplicants wake in a cramped escape pod with a single mandate — make air, make food, make water, and survive the first hundred cycles. This chapter teaches the whole first breath: how to control the game, read the interface, and keep three newbies alive until the colony runs itself. Everything here describes vanilla Oxygen Not Included; Wave 0 installs no content mods, and the few mods in the Modlist chapter only make the game easier to see, operate, or plan.

The First Breathe's Roadmap

Before the keybindings, here is the whole wave in one glance — four milestones, each building on the last:

1. **Survive the first day.** Pause, dig a little breathing room, put down an outhouse, a wash basin and a cot (**Your First Minutes**).
2. **Stabilize air, food and water.** An **Oxygen Diffuser** burning algae, a small **mealwood** farm, and a plumbing loop that recycles polluted water (**Oxygen & Water Production**, **Food & Farming**, **Plumbing & Sanitation**).
3. **Earn your first research.** A **Research Station** turns dupe labor into science points, and science unlocks every machine above (**Research**).
4. **Ride out the hundred cycles.** Power without a dupe on the wheel, priorities that keep everyone busy, a hatch ranch feeding the colony and the coal generator (**Power & the Electric Grid**, **Duplicants**, **Critters**).

Winning Wave 0 means **stable**, not thriving: oxygen holding, calories climbing, water cycling, no one in the med bay. When those systems run on their own, the hundred cycles are earned — and every later wave just makes that stability bigger.

Controls & Keybindings

Oxygen Not Included is a colony simulator: you do not steer a character, you command a colony — choosing a tool, pointing it at the map, and letting the duplicants work. The defaults below are what the game ships with; everything is remappable in the settings.

Action	Input
Pan the camera	WASD or arrow keys
Zoom	Mouse wheel
Pause / unpause	Space
Game speed	1 / 2 / 3 (or the on-screen buttons)
Open the build menu	Right-click, or the build toolbar at the bottom
Priority overlay	P
Cancel the current tool	Esc

The mouse does the rest. Left-click selects or inspects, right-click cancels, and clicking and dragging on the map **drag-selects** a group of duplicants to command at once.

Digging and building. The bottom build menu holds every tool and building by category — Oxygen, Plumbing, Power, Food. Select **Dig** and click or drag to mark excavation; select a building and click to place, or drag to queue several. Green means the job can be done now; red means blocked, and it waits until the obstruction clears.

Priorities. Press **P** for the priority overlay, then click or drag over tasks and buildings to set them 1 (ignored) to 9 (urgent) — higher numbers get worked first, exactly what you need when the oxygen is running out.

Tip: Space is the most important key in the game. Pause, read the alerts, queue a handful of orders, then unpause and watch. New players who refuse to pause are always one disaster behind.

The Interface Tour

The whole HUD is a few panels, and each answers one question:

- **The top bar** is your colony's vital signs: the cycle clock, and a portrait strip of your duplicants — each portrait shows that dupe's oxygen, calories and stress. When someone is suffocating, starving or about to snap, it tells you first.
- **The build menu (bottom)** holds every tool and building by category; the **Dig** and **Mop** tools sit here too.
- **Speed and pause**: pause, then 1×, 2×, 3×. Pausing freely is the first skill — the game only moves when you let it.
- **Overlays**: icons that tint the map — light, heat, electricity, plumbing. The electrical overlay helps when something stops drawing power.
- **Notifications**: a ticker of events — "Wire overloaded", "Outhouse full", "Slimelung detected". Click an alert to jump to the problem; treat them as a to-do list.
- **Top-right tabs**: **Research** (flask) shows the tech tree; **Jobs** assigns roles; **Duplicants** manages priorities and skill points; the schedule clock splits the day into work, sleep and downtime.

Hovering anything — a building, a resource, a research card — shows a tooltip explaining what it is and what it needs. The game's documentation lives there; read it whenever a building surprises you.

Your First Minutes

The opening checklist, in order. Do not rush it — pause, do each step, check the result.

1. **Pause.** Hit Space and look at what you have: three duplicants, the pod, a small pocket of air.
2. **Dig a little breathing room.** The pod pocket is cramped — mark a modest chamber, not a mine in every direction.
3. **Build an outhouse and a wash basin.** Put them close together and close to where the dupes work. Both need periodic emptying — the outhouse into a **Compost**, the basin by hand — because a full outhouse stops working and dupes start using the floor.
4. **Set up sleeping.** Build a **Cot** for each dupe in a quiet corner. Floor-sleepers wake with a sore back, and tired dupes work slower and stress faster.
5. **Set priorities.** Open the **Duplicants** tab and split the tasks — one digger-heavy, one builder, one researcher — so they are not all digging while nobody researches.
6. **Start research.** Build a **Research Station**, assign a dupe with research priority, and queue **Basic Farming** first so planter boxes unlock (see **Research**).
7. **Unpause and watch.** Let the first jobs complete. Whatever the alerts complain about next is your next job.

Oxygen & Water Production

Air is the first problem and the one that never goes away. The pod's pocket is breathable, and the rocks you dig up hold oxygen too — but three duplicants breathe it down fast.

The Oxygen Diffuser. Build one, connect power, and feed it algae — it converts algae into oxygen, your first real oxygen source and the workhorse of the whole wave. It needs **algae**, a sandy-looking resource in the starting biome, and it gives off a little heat while it runs.

> **Warning:** Algae looks plentiful and is not. The Oxygen Diffuser eats it steadily, and a base that runs out of algae suffocates quietly. Watch the stockpile in the top bar's resource panel and treat a falling algae line as an emergency, not a curiosity.

Carbon dioxide pools at the bottom. CO_2 is heavier than oxygen, so it sinks into the lowest spots of your base and slowly suffocates anyone working low. Keep the low points open so CO_2 has room to pool harmlessly.

Polluted oxygen. Polluted water and polluted dirt off-gas **polluted oxygen**: foul but breathable, and a sign of decay. A little is tolerable, but it carries germs — do not let it become the colony's air supply.

Water. Your starting supply is the clean pool in the starting biome. Dupes drink from open water, and if the only nearby pool is polluted they drink that and get food poisoning — keep polluted water away from where they drink. The **Water Sieve** later (see **Plumbing & Sanitation**) turns it back into clean water.

Electrolyzers. A larger machine splits water into oxygen and hydrogen and makes a **lot** of air — but it also makes hydrogen you must manage and draws real power. That is Wave 1's topic; for the first hundred cycles, algae is your air.

Food & Farming

Calories come next. Your pod starts with a small ration of **meal lice**, enough for a few days — your buffer, not your plan.

Meal lice and mealwood. Meal lice is the raw food; **mealwood** is the plant. Unlock the **Planter Box** (Basic Farming research), build a small farm, plant mealwood, keep it watered, and it steadily produces meal lice. A handful of plants feeds three dupes.

Cooking. The **Electric Grill** cooks raw food into better food: **Pickled Meal** from meal lice keeps far longer than raw — the difference between food that keeps spoiling and food you can stockpile. The **Microbe Musher** mixes meal lice and water into **Liceloaf**, a better meal, once you have water to spare.

Spoilage. Food rots over time; raw spoils quickly, cooked and pickled lasts much longer. Store food in a **Ration Box** early, later in a powered **Refrigerator**. Rotten food is useless, so eat the old food first.

Ranching. Later, farming grows a second leg: hatches, covered in **Critters: Hatch Ranching Basics**. For the first sessions, plant mealwood and cook.

Power & the Electric Grid

Almost everything that keeps the colony alive needs watts, and watts do not appear by themselves.

The hamster wheel. The **Manual Generator** produces power while a dupe runs it — labor. Every watt costs a dupe who could be digging or researching; fine for a light or a pump, a treadmill trap for a whole base.

The coal generator. The **Coal Generator** burns coal — which you dig up everywhere — for steady, dupe-free power, and it is the wave's main generator. It produces heat and carbon dioxide, so give it a ventilated spot and keep a coal stockpile beside it.

Wires and capacity. Every circuit has a limit. The standard wire overloads at roughly 1,000 watts of simultaneous draw — cross that and the wire takes damage, shorts, and can burn the circuit out, machines included. Spread high-draw buildings across separate circuits, or upgrade the wire when you have the research.

> **Warning:** A wire that overloads takes damage and can burn, dropping the whole circuit — and the Oxygen Diffuser on it — at the worst possible moment. Open the electrical overlay (the circuit icon), and when a circuit shows more than its rated load, split the consumers before the wire sparks.

Batteries. Generators produce in bursts; **batteries** (the **Small Battery** early, the **Jumbo Battery** soon after) store the surplus so a dip in generation does not dip the colony's power.

Why power matters. Pumps, the diffuser, the grill, later research machines — every self-sufficient system needs electricity. When the colony stalls and nothing seems broken, check the power first: a dead wire, an empty coal pile, a dupe who stopped running the wheel.

Plumbing & Sanitation

Sanitation is the difference between a colony and a plague house — and the first place liquid **pipes** earn their keep.

Outhouses and wash basins. The outhouse needs no water: a dupe uses it, and a dupe empties it into a **Compost** when full. The wash basin — right by the outhouse door — is where dupes wash; it collects polluted water a dupe empties by hand. This unplumbed pair keeps early germs down with zero pipes and power; keep the emptying jobs on someone's priority list.

Pipes. Later you graduate to **Liquid Pipes**: clean water in, polluted water out. Pipes are where plumbing goes wrong — a leak or dead end quietly floods a floor, and dupes wading in polluted water get sick. Check the plumbing overlay when something floods.

The sieve. The **Water Sieve** closes the loop: feed it polluted water plus a filtration medium — sand is perfect, and you dig up sand constantly — and it returns clean water, plus a little polluted dirt. It turns a sewage problem into a water supply.

The classic bathroom loop. Build a **Lavatory** (a plumbed toilet): pipe clean water in, the polluted water out to the sieve, and the sieve's clean output back. One catch: a lavatory puts out a little more polluted water than it takes in, so the loop slowly makes a surplus — pipe it into a **Liquid Reservoir** near the farm or the base will flood, and never let the loop dead-end.

Don't drown your dupes. Plan the whole loop before you place a single pipe — the source, the pump (if any), the sieve, and somewhere for the surplus to go. A pump with no pipe does nothing, and a pipe with no destination floods.

Duplicants: Needs, Priorities, Skill Points & Morale

The colony is three individuals with opinions, and managing them is half the game.

What a dupe needs. Every duplicant needs, in rough order of urgency: **oxygen** (they hold their breath a while, then suffocate), **calories** (they eat a couple of times a day), **sleep** (a cot, not the floor), a **bathroom** (a desperate dupe uses the nearest water — including a polluted pool, which poisons them), and **comfort**: acceptable temperature, some decor, time off. Unmet needs push **stress** up, and high stress makes dupes vomit, melt down, and wreck the base. Watch the stress bars the way you watch the oxygen.

Priorities. Priorities exist at two levels. In the **Duplicants** tab, each dupe ranks task types — Digging, Building, Researching, Operating — deciding what they prefer. On the map, the **priority overlay** (P) sets how urgent a specific task is. A dupe with Researching maxed and a Research Station marked priority 9 will be at the lab before anyone digs. The beginner trap is setting everything to 9 — when everything is urgent, nothing is.

Jobs. The **Jobs** tab turns skill points into roles — Rancher, Digger, Miner. Assigning a role costs skill points and raises the dupe's **morale expectation**, but it is how three generalists become three specialists.

Skill points. Dupes earn experience from working and gain a **skill point** when they level up. Spend them where their priorities already point — a dupe who never researches is a wasted researcher.
Morale. Morale is a dupe's contentment, raised by decent food, decor, a comfortable bed, and downtime. The catch: every skill point spent raises the dupe's **morale expectation**, and when expectation climbs above morale, stress climbs with it. Raise morale first — better food, a painting, a proper schedule — then spend the points.

Tip: Treat morale as a budget. Before you spend a dupe's skill point, check that their morale can cover the higher expectation — spending points faster than you can raise morale is the fastest route to a stressed, vomiting colony.

The schedule. The schedule tab splits the day into **work**, **sleep**, and **downtime** — when dupes eat and relax. Too little downtime and morale falls and meals get skipped; a generous block beats a maximum-work schedule that burns everyone out.

Critters: Hatch Ranching Basics

The starting biome is home to **hatches** — plodding, harmless, round rock-eaters — your gateway to ranching and the only ranch you need in Wave 0.
What they eat. Hatches eat raw minerals: the **sedimentary rock**, **regolith** and other loose rock you are digging up anyway. Your trash is their dinner.
What they give. A fed hatch produces **coal** (their waste, useful once you build a coal generator), **eggs**, and eventually **meat**. A ranch turns the rocks you mine into fuel and food at once.
Taming. Wild hatches become **tame** when fed over time, and tame hatches stay put, breed, and lay eggs reliably. The **Critter Drop-Off** rounds them into an enclosed room — the **stable** — so they cannot wander off.
The ranch loop. Put a **Critter Drop-Off** and a **Grooming Station** in the stable and assign a dupe the **Rancher** job. A groomed, tame population slowly grows; eggs hatch into more hatches or get cracked and cooked into omelettes. The coal feeds the generator, the meat feeds the colony — rocks in, fuel and food out.
Start with one or two hatches and a small stable — the only system in Wave 0 that pays back in three resources at once.

Research

Everything you are allowed to build is locked behind **research**, and at this wave research costs time and dupe labor, not resources.
The Research Station. Build it and a dupe operating it steadily produces **research points**. It needs no power — just a dupe who is not busy dying. One station and one researcher is the baseline; a second station doubles the pace.
The early research order. Pick what keeps you alive before what makes you comfortable:

1. **Basic Farming** — the Planter Box and mealwood, your food security.
2. **Plumbing** — liquid pipes and the **Water Sieve**, turning bathrooms and pollution into a water loop.
3. **Oxygen** — the **Oxygen Diffuser**, your first real oxygen production.
4. **Interior Decor** — paintings and plants, better morale for everyone.
5. **Power** — the **Coal Generator** and batteries, power without a dupe on a wheel.

A suggestion, not a law — every asteroid is different — but food, air, and water always beat cosmetics.
Why research speed matters. Every machine in this chapter sits behind one of those researches. A colony with an idle Research Station — no dupe assigned, or the researcher drafted for digging —

stalls even with plenty of algae and coal. Once the basics are built, keep someone researching most of the time.

Common Beginner Mistakes

- **Heat creep.** Every machine — the diffuser, generators, the grill — bleeds heat into the base. A base that starts pleasant slowly creeps toward crop-killing heat if it is never considered. Do not panic in the first fifty cycles; do not ignore it either.
- **Running out of algae.** The diffuser is hungry and algae runs out faster than it looks. Mine it whenever you see it.
- **CO2 pockets.** Digging down without a plan traps your dupes in suffocating low pockets. Keep low rooms open so the CO2 settles harmlessly.
- **Over-expanding.** A big base needs light, power, cooling and dupes to maintain it. Grow one room at a time instead of mining in every direction.
- **Dupes sleeping on the floor.** Cots are nearly free; a floor-sleeper starts every day exhausted and stresses faster.
- **Ignoring morale.** Spending skill points while morale is low is the stress spiral. Keep the seesaw balanced.
- **Digging straight down into hazards.** Below the start waits the slime biome — slimelung germs, hot pockets, buried water. Dig down slowly and with an escape route.
- **Building pumps before understanding them.** A pump does nothing until it has power, a pipe to push into, and somewhere to send the liquid. Plan the whole line first.

None are fatal, and Wave 0 expects you to make all of them — that is what the hundred cycles are for. Fix each one as the alerts name it, and the colony gets stronger every time.

The First Breathe's First Session

If you want to close the chapter with your hands on the keyboard, here is one 15–20 minute sitting that touches every system and leaves the colony able to breathe on its own:

- **Minute 0–5:** pause. Dig a small chamber around the pod, place an **outhouse** and **wash basin** side by side, and build a **Cot** per dupe in the corner (**Your First Minutes**).
- **Minute 5–10:** build a **Research Station**, set one dupe's research priority high, and queue **Basic Farming**. Unpause and start digging (**Research**, **Duplicants**).
- **Minute 10–15:** once the Planter Box unlocks, build a small mealwood farm near your water and plant it. Build a **Manual Generator** and a **Small Battery** so you have a circuit ready to power the Oxygen Diffuser you are about to build (**Food & Farming**, **Power & the Electric Grid**).
- **Minute 15–20:** research **Oxygen** and build an **Oxygen Diffuser** on the battery circuit. Feed it the algae you mined, keep research moving with a second dupe, and empty the outhouse when the alert appears (**Oxygen & Water Production**, **Plumbing & Sanitation**, **Duplicants**).

Checkpoint: the colony can breathe on its own — the diffuser runs, the research station is busy, food is growing, the outhouse loop is staffed. That is the first stable breath, and the definition of winning Wave 0's first session. Stop there; everything after is the same loop, just bigger.

Wave 0: Modlist

Wave 0 adds no gameplay content. Every mod in this chapter exists for one of three reasons: to make the game easier to **see**, easier to **operate**, or easier to **plan**. Nothing here changes a recipe, a resource, a building stat, or a research tree — that is the wave's hard rule. Wave 0 admits only **CLIENT-ONLY** mods (purely visual, UI, or informational; they never touch the simulation) and **BEHAVIORAL-NEUTRAL** mods (they change how you interact with buildings — copy settings, plan layouts, filter sweeps — without changing any value the simulation uses). Anything that alters a stat, a recipe, or a balance number is Wave 1+ material and must clear the power-spike gate first.

As of this release, every card below is **pending addition**: per project decision, this project does not research or select mods — you do. When you are ready, paste one verified mod card per mod into the placeholder blocks in the Mechanics and Graphics sections that follow. Each card needs a real Workshop URL, a classification tag, and a power-spike pass, per the rules in the project's STATUS.

The mods you add split across the next two sections by what they change: **Mechanics** for tools that change how you work, and **Graphics** for mods that only change how the game looks and how it tells you what is happening.

Wave 0: Mechanics

Mechanics mods change how you **interact** with the colony, not how the colony behaves. For a quality-of-life wave that means the tools of comfortable operation: larger and more readable menus, copy-settings helpers that clone one building's configuration onto a row, planning aids that make layouts easier to think about, and sweep or selection filters that keep the chore of tidying the base fast. None of these touch a building stat, a recipe, or a balance value — they make the game easier to run, not easier to cheat.

Wave 0: Graphics

Graphics mods change only how the game looks and how it tells you what is happening — nothing about the simulation underneath. Expect cleaner overlays, color-coding that makes statuses readable at a glance, and clearer or quieter notifications. A mod in this section should make the colony easier to read, never easier to cheat.

Wave 1: The Vent Tamer

Wave 1: Strategy

Wave 0 bought the colony another hundred cycles by learning to stand still. Wave 1 is the wave that starts moving: the mandate is to tame the world — box the geysers, answer the heat, turn the oil below into power and plastic, and make the base fast enough to ignore. Every section below assumes the Wave 0 basics are stable, and every one hands you a system you will run for the rest of the colony's life.

Automation & Logic Circuits

The wires you built in Wave 0 carried power; a second kind of wire — the **automation wire** — carries **decisions**. Automation lets buildings talk to each other: a sensor says "the room is full", a shutoff valve reads that and closes, a generator reads its battery and powers down. It is the difference between a colony that is merely alive and one that runs itself.

Signals. Automation wires carry one of two states — **green** (on) or **red** (off). A sensor outputs green when its condition is true: a **Liquid Pipe Element Sensor** when the pipe holds water, a **Gas Pipe Element Sensor** when the pipe holds a chosen gas, a **Hydro Sensor** when the liquid in its cell is above a depth you set. Whatever a sensor decides is sent down the wire to whatever the wire touches.

Reading it. The automation overlay shows every wire green or red, so the loop's state is visible at a glance — most of "automation" is really just getting this overlay right.

Actuators. Wires only inform; **actuators** act. The two you need first are the **Liquid Shutoff** and **Gas Shutoff** — valves with an automation port that open when the wire is green and close when it is red. A shutoff does not pump; it gates whatever the pump already pushes, so pair it with a pump that runs continuously and let the shutoff decide where the liquid goes.

Logic. When a sensor's answer is backwards — you want to act when it is **red** — the **NOT gate** flips it. A hydro sensor says "too low" with a red wire; feed that into a NOT gate and the wire comes out green exactly when you want a pump to run. NOT is the first logic gate and the only one you need for most of Wave 1.

A first build: the smart battery. The **Smart Battery** is a battery with an automation output: it sends green when its charge drops below a low threshold and red when it climbs past a high one — both numbers you set on the battery itself. Wire that output to a **Coal Generator** with an automation port, set the battery to charge to, say, 90% and release at 40%, and the generator now runs only when the battery is genuinely low. When the battery is full the generator stops, the dupes stop hauling coal, and the coal pile stops evaporating. It is the single highest-value automation build in the game, and the template for every later loop: **sensor reads, wire reports, machine reacts.**

> **Tip:** The smart battery pair — generator stops when the battery is full, restarts when it is nearly empty — is the colony's first self-running system. Build it before anything fancier: it saves coal, heat and dupe labor from the moment it is wired.

Once the smart battery works, you have the whole vocabulary — sensor, wire, shutoff, NOT. Every tamer in the rest of this wave is the same parts arranged differently.

Geyser & Vent Taming

Wave 0 dug for its water and gas. Geysers are the map's **renewable** sources: vents and geysers buried in the asteroid that periodically erupt a fixed resource — cool or hot water, steam, natural gas, chlorine, carbon dioxide, polluted water, hydrogen. They are the reason "survive" becomes "tame": a tapped water geyser feeds the colony forever, a natural gas vent runs generators, a steam vent is a power plant waiting for a turbine. The map marks them as blocked-off rock formations; each hides a machine that erupts on a schedule of active and dormant periods.

Why they matter. A geyser is free mass over time — one of the only sources in the game that adds water, gas or oil from nothing. But it all comes with a price: heat. Most erupt hot — sometimes very hot — and a geyser left in the open turns its surroundings into a slow oven. Taming a geyser means doing two things: **boxing it in** so its output stays put, and **pumping its output away** faster than the pocket fills.

Overpressure. Every vent stops erupting once the pressure in its chamber climbs high enough — a geyser buried in its own output just shuts down. That is not a bug; it is the mechanic you exploit. A vent kept at low pressure erupts continuously; a vent allowed to back up goes dormant on its own schedule. So the goal is never "let it fill" but "pump it out steadily and keep the pressure low."

The beginner tamer. The first tame a Wave 1 colony should build:

1. **Box it.** Wall the geyser into a chamber with insulated tiles and a space to stand. Sealed is fine; vacuum is an advanced refinement, not a requirement.
2. **Pump the bottom.** Put a **Liquid Pump** in the pocket where the geyser's output collects. (Gas vents take a **Gas Pump** instead.)
3. **Gate the pump.** Run the pump continuously, but send its output through a **Liquid Shutoff**, and put a **Hydro Sensor** in the pocket — wired through a NOT gate if you want it inverted — so the shutoff opens only when enough liquid has pooled. A pump that runs dry wastes power; a pump that runs only when there is water wastes nothing.
4. **Send it somewhere.** Pipe the output to a **Liquid Reservoir** or straight into the base's water line. The reservoir also breaks the loop — a pipe with no destination floods the chamber and re-wets everything.

> **Warning:** Never tap a geyser you are not ready for heat-wise. A hot steam vent left unboxed cooks everything within a dozen tiles — box it in insulated tiles the same day you open it, and plan where its heat is going before you decide it is safe.

Heat Management & Steam Turbines

Wave 0 warned that heat slowly creeps toward crop-killing temperatures and told you not to panic. Wave 1 is where you actually answer it, because by now every machine — diffusers, generators, refineries, and the geysers you just tamed — is bleeding heat into the base, and the base slowly, quietly cooks. Crops wither at high temperatures. Dupes take stress damage in hot rooms. And heat never disappears by itself: in ONI, heat only leaves if you **delete** it with a machine built for the job.

Insulate first. The cheapest defense is the **Insulated Tile** — a wall that lets almost no heat through. Ring your living and farming rooms with insulated tiles, and keep hot industry — generators, refineries, geyser chambers — outside the ring. Most Wave 1 heat problems are solved by walls and spacing, not machines.

The aquatuner. The **Aquatuner** is a machine that cools the liquid piped through it: water in, noticeably colder water out, fed back through your base's pipes. It is the closest thing the game has to a cooling plant for liquid loops — with one enormous catch. The heat it pulls out of the liquid does not vanish. It is dumped straight into the machine itself, and an unmanaged aquatuner will melt its own room in a few cycles.

The steam turbine. The **Steam Turbine** is the machine that answers the aquatuner's catch. Built above a sealed chamber, it pulls in hot steam, converts its heat to power, and spits out the leftover as cooler water. A steam chamber feeding the turbine is a **heat deletion machine** — the real one in the base game: heat goes into the chamber as steam, the turbine consumes it, and the mass leaves cooler than it arrived. That is not a balance quirk; it is the intended endpoint of every cooling loop.

The loop. Put them together and you have the colony's heat sink: an **Aquatuner** in a sealed steam chamber, chilling a pipe of liquid that circulates around your base; the heat it pulls dumps into the chamber's water and turns it to steam; the **Steam Turbine** on top eats the steam, returns a little power, and drops the condensed water back into the chamber to be reboiled. The aquatuner cools the base; the turbine pays back the power and closes the loop. It runs forever so long as the chamber stays fed.

Gas heat. What cools a liquid can cool a gas: the **Thermo Regulator** works exactly like an aquatuner but for gas pipes — same principle, same heat-dump problem, and the same steam-chamber solution once you need it. Use it for gas loops; use the aquatuner for everything liquid.

> **Note:** The exact temperature numbers — how cold the aquatuner makes the water, how hot the steam must be — shift with the game's tuning and with what you are cooling. Design the loop as **an idea**: the aquatuner pulls heat out of a loop, the chamber absorbs it, the turbine deletes it. The numbers are the tuning; the loop is the truth.

Plastic & Petroleum

Deep below the slime biome sits the **oil biome** — a hot, dark, unwelcoming place where the rock runs black and cold stone is scarce. This is where Wave 1's fuel economy is decided, because this biome holds the oil that becomes petroleum and, through it, plastic.

Oil wells. Oil in the ground is effectively endless, but it is guarded: an **Oil Well** drills the oil reservoir and requires **water** to do it — pump clean water in, crude oil comes out, with a little natural gas as a byproduct. An oil well is how you turn a water surplus into fuel; it is also how you can empty a water reserve without noticing. Watch the water line the way you watched the algae line in Wave 0.

The oil refinery. Crude oil is a poor fuel. The **Oil Refinery** cooks it down into **petroleum**, a better fuel, and vents natural gas while it works. A refinery needs a dupe at the controls, produces real heat, and — like the geyser tamer — needs its output piped away and its heat contained. One refinery turning crude into petroleum is the standard Wave 1 setup; scale up when demand shows up.

Petroleum generators. The **Petroleum Generator** burns petroleum for the most power per tile in the wave — but it is loud about it: it needs a steady fuel supply, produces waste that must go somewhere, and generates a lot of heat. It is the generator you upgrade the coal fleet into once the oil line is stable, not the first generator you build.

The polymer press. The **Polymer Press** is the building that turns petroleum into **plastic** — and the same machine turns its water feed into steam, so it also needs a heat plan. Plastic is the wave's keystone material: it is required to build the **Steam Turbine** from the previous section, and it unlocks the **Transit Tubes** at the end of this chapter. It is also a research gate — most of the mid-game tech tree expects you to have plastic in hand.

What plastic is for. In Wave 1, two things: the steam turbine, and transit tubes. Plastic is fragile — it melts in heat — so keep your plastic stockpile cool and out of industrial rooms.

> **Warning:** The oil biome is hot and its industry is hotter. Do not build your oil refinery inside your insulated core — build it in the oil biome or just outside the ring, and treat its heat as a geyser you are taming, not a room you are sharing.

Ranching Diversity

Wave 0 ranched hatches, and it is a good start — rocks in, coal and meat out. Wave 1 widens the menagerie, because each new critter is a machine that turns one resource you have too much of into one you need.

Dreckos. Dreckos are tree-dwelling reptiles that eat **mealwood** (and later **balm lily**) and are farmed for what they grow, not what they drop: shear a drecko and it yields **reed fiber**, the fabric for insulation and exosuits, alongside **phosphorite**, a fertilizer. A drecko ranch turns a plant surplus into clothing material and fertilizer.

Pufts. Pufts are the pollution recyclers. They eat **polluted oxygen** — the foul air your polluted water and dirt keep generating — and excrete **slime**, which refines into algae and, eventually, mushroom farming. A puft ranch converts a nuisance (polluted oxygen seeping everywhere) into a resource (slime you actually want).

Slicksters. Slicksters are the oil biome's answer to CO_2. They eat **carbon dioxide** — the gas your dupes and coal generators never stop producing — and excrete **crude oil**. Park a few in the colony's CO_2 pit and the bottom of the base becomes a slow oil well. (A hotter variant eats CO_2 hotter and leaves petroleum instead; that is a Wave 2 refinement.)

Shine bugs. Shine bugs are the glow critters of the starting biome: they eat **algae** and emit **light** — bright, warm, cheerful light. A shine bug or two in a living room is free lighting and free morale, as long as you keep feeding them.

The principle: close the loop. Every critter ranch above is one half of a loop looking for its other half. Hatches eat rock and feed the coal generator. Slicksters eat the generator's CO_2 and return oil. Pufts eat the oil-era pollution and return slime. Shine bugs eat algae and brighten morale. The ranching skill is spotting the waste stream and matching a critter to it — the colony stops throwing things away and starts digesting them.

Dupe Jobs, Skill Specialization & Morale Optimization

By Wave 1 the colony has enough dupes that "everyone does a bit of everything" is how nothing gets done. The **Jobs** tab is where three generalists become a crew: each job path is a ladder of **skill tiers**, and spending skill points down a path unlocks that tier's abilities — **Hard Digging** so a digger can excavate hard rock, the **Plumbing** skill so a builder can lay pipes faster, the **Improved Carrying** skills that let one dupe do the work of two haulers.

Specialize on purpose. A colony this size runs on a handful of specialists: one dedicated **researcher** who lives at the research station, one **builder** with the construction skills, one **digger** with the digging skills, one **rancher** for the stables, and everyone else set to operating and hauling. Specialization is not about the titles — it is about the **skill points** being spent in one place instead of sprinkled everywhere. A dupe with five points in digging is worth more than five dupes with one point each.

Morale is the budget. Every skill point spent raises that dupe's **morale expectation** — the contentment they now consider baseline. Their actual **morale** comes from food quality, decor, sleep, downtime, and a few building bonuses. The rule is a budget: expectation must stay under morale, or the dupe stresses, vomits, and eventually melts down. So every promotion is a bill, and the bill is paid in food and decor.

The morale crash spiral. The classic collapse: you promote a dupe (expectation up), then another, then run out of decent food — and suddenly three dupes are stressed, one is having a meltdown, and the whole base pauses to mop vomit. The spiral feeds itself because stressed dupes work poorly, which delays the food or decor that would have fixed it. The defense is boring and reliable: **raise morale before you spend the points.** A **Mess Table** and a proper mess hall, a **Painting** or two, a decent **Cot**, and a schedule with real downtime all raise morale for everyone — upgrade comfort first, then promote.

Warning: Promotions are permanent spending, not rewards. If a dupe's morale expectation is already at the ceiling of what you can feed and decorate, adding one more skill point is how you start the stress spiral. Look at the two numbers side by side before every promotion.

Transit Tubes

By the end of Wave 1, the base is big enough that walking is the real cost — a dupe trekking across the asteroid is a dupe not building. **Transit Tubes** are the answer: a plastic-based transport network that carries dupes across the map almost instantly.

How they work. A transit network is built from three pieces: the **Transit Tube Access** (the launch pad a dupe enters), the **Transit Tube** itself (the pipe the dupe is shot through), and a **Transit Tube Crossing** where tubes intersect. The launching access needs power; the rest of the tube just needs plastic and a clear straight line. A dupe steps in, is launched, and exits at the nearest access along the tube — effectively teleporting between your base and your far industry or geyser. Tubes are strictly one-way in spirit: plan a line for each journey you actually want, from the ranch to the refinery, from the base core to the geyser field, and put an access at both ends so the trip works in both directions.

What they cost. Tubes are pure plastic — a cross-map line burns through your entire early polymer press output — and the access buildings need power at each launch point. They are a material investment, not a science curiosity.

When they are worth it. The rule of thumb: build tubes when the **walk** is the bottleneck. A colony with its ranch, oil refinery and geyser tamer all in one compact area does not need them; a colony stretching across a third of the map, with dupes spending a quarter of every day walking, is exactly the colony tubes are for. Transit tubes are also the game's polite nudge toward the next wave — they are the first taste of travel technology, and plastic is the toll.

Tip: Price the tube line honestly before building. One long line of plastic is a big early spend; if the map is still compact, the same plastic as a second Polymer Press may serve better. When the base gets wide, tubes pay for themselves in dupe-hours within a few cycles.

Wave 1: Modlist

Wave 1 is where the colony stops surviving and starts engineering, and the mods in this chapter are allowed to change how the game plays — not how the game cheats. Where Wave 0 admitted only mods that changed nothing about the simulation, Wave 1 opens the door to **STAT-CHANGING** mods: mods that alter a recipe, a resource, a building's numbers, or how a system works. That door has one lock on it — the **power-spike gate**. A Wave 1 mod passes only if it does not hand the colony free resources, free heat deletion, or free power — nothing that lets a player skip the taming work this wave is about. Mods that reveal geysers, drain heat for nothing, spawn water or oil from empty rooms, or remove the cost of an important build are out, because they delete the very game this wave teaches.

As of this release, every card below is **pending addition**: per project decision, this project does not research or select mods — you do. When you are ready, paste one verified mod card per mod into the placeholder blocks below, with its Workshop URL, its classification, and a note recording its power-spike verdict.

The mods you add split across three sections by what they change: **Mechanics** for automation, thermal, and geyser tools; **Content** for new buildings, critters, plants, and recipes; and **Graphics** for overlay and visual clarity.

Wave 1: Mechanics

Mechanics mods change how the **systems** behave — the automation that runs your loops, the heat transfer your base lives with, the behavior of the geysers and vents you tame. These are the first Wave 1 mods allowed to touch simulation values, so every one must clear the power-spike gate: no free resources, no free heat deletion, no skipped work. A mechanics mod should make taming deeper or clearer, never cheaper.

Wave 1: Content

Content mods add things to build, ranch, grow, or cook — new buildings, new critters and plants, new recipes. The standing rule for Wave 1 is **proportionate cost**: anything new must cost what it is worth, so a powerful building or critter comes with a matching price in materials, power, or labor. A content mod that hands out a cheap version of something expensive fails the power-spike gate.

Wave 1: Graphics

Graphics mods change only how the game looks and how it tells you what is happening — the overlays you read while taming vents, the clarity of the temperature and gas views, the legibility of the whole screen. They never touch simulation values, so they carry no power-spike risk. What they must do is stay honest: clearer, not smarter — a geyser is still a geyser until you tame it yourself.

Wave 2: Strategy

Wave 0 taught a pod to breathe. Wave 1 taught a base to run itself. Wave 2 is the wave that leaves: the colony has outgrown its starting asteroid, and the mandate is to cross the void, claim new worlds, and finish the Spaced Out endgame. Everything below assumes Waves 0 and 1 are stable, and assumes the **Spaced Out!** expansion — the rocket platform, the star map, the other planetoids. Where Wave 1 tamed a single world, Wave 2 teaches you to run a network of them, and every section is a system you will fly, ship, or land until the last planetoid is settled.

Rocketry

In Spaced Out, a rocket is not a vehicle in the sci-fi sense — it is a colony you strap an engine to. Rockets are built on the **Rocket Platform**, a pad you build on the surface, and they are stacks of modules: engine at the bottom, nosecone or capsule at the top, and everything between is the rocket's **interior** — pressurized space that dupes can stand in, build in, and live in while the rocket travels. The rocket is the only way off your starting asteroid in Spaced Out; the star map is reachable only by launching, and the rocket is both the ship and the base you land inside.

The modules define what the stack can do. The **Command Module** is the pilot's seat — a rocket needs one piloted seat to fly under control. The **Trailblazer Module** carries a one-way landing pod that drops a scout onto a planetoid before any landing pad exists there — the first-contact tool. The **Drillcone** sits at the front of the stack and carves through the space debris and asteroid crust that otherwise block the route; a rocket headed to a new world generally wants one. **Cargo bays** carry freight — but in Spaced Out a cargo bay is also **building space**, interior a dupe can fill with whatever the new colony needs. Because interiors are buildable, the rocket is not just transport: it is a moving base, and the modules you stack decide whether it arrives as a mail crate or a small house.

Fuel decides how far the stack can go. Early rockets burn **petroleum** — the refined oil from Wave 1 — which is cheap and reaches the closer worlds. The endgame fuel is **liquid hydrogen**, far more efficient per kilogram and the practical choice for the long routes to the edges of the star map. Moving from one to the other is a real project: hydrogen liquefaction demands the Wave 1 cooling loops pushed to cryogenic temperatures, and the fuel infrastructure — production, storage, refueling — becomes its own industrial corner of the base.

One design decision frames everything above: **solo nosecone or full interior.** The **Solo Spacefarer Nosecone** turns a rocket into a single-seat capsule — one dupe, no interior to build, the cheapest way to hop a pilot around. A full interior costs more: more modules, more mass, more fuel to lift the whole stack — but it carries a work crew and everything they need to land and build. Most mature colonies run both kinds, the small capsules for scouting and supply runs, the large interiors for colonization.

> **Note:** A rocket platform needs open sky. A ceiling, a ledge, or a parked rocket overhead blocks launches, so clear the surface before you build and leave room around the pad for the pipes, wires and rails the stack will connect to.

Interplanetary Logistics

Wave 0 and 1 taught one mental model: one big base, every resource inside a single growing shell. Spaced Out breaks the model on purpose. The starting asteroid cannot produce everything — or at least not enough of it. Some planetoids hold oil where yours has none; others sit on uranium; a water world is practically made of the stuff. The colony's future is not "expand the base" but "connect the bases." You stop running a colony and start running a **network**: a handful of outposts, each producing what it is good at, each importing what it cannot make.

Resources move between worlds by two routes. The first is **cargo modules** riding a rocket: load the bay, fly, unload at the far pad — heavy, able to carry anything, but it needs a pilot and a flight every time. The second is the **Interplanetary Launcher**: a fixed building that flings a payload across the star map to a landing pad on another planetoid, no crew required. The launcher trades the ability to carry a colony for hands-off convenience — it runs on power and keeps launching as long as it is fed and aimed. Most networks use both: rockets for the big scheduled lifts that start a colony, launchers for the steady daily trickle that keeps it alive.

Ship the finished good, not the raw ore. The Wave 1 lesson — refine, cook, and craft where the work is done — applies across the network: moving plastic from the base that makes it beats shipping petroleum and running a second press on a water-starved moon. The further the destination, the more valuable every kilogram of payload, and a launcher pod is a poor place for a crate of ore you could have processed at home.

Launch costs make automation inevitable. Every flight burns fuel and crew attention, and a rocket flown half-empty is a rocket that wasted its toll. The moment the same route repeats every few cycles, automate it — a scheduled launcher feed, a standing cargo order, a rocket that is loaded and dispatched without a dupe deciding. A network that depends on you hand-flying each shipment is a network that will starve the moment you are looking at the wrong planetoid.

Tip: Design every outpost with its own basic life support before you import anything. A colony that must ship oxygen to stay alive is always one missed launch from suffocating — freight should be for growth, not for breathing.

Planetoid Colonization

The star map around you is a menu of decisions, and each planetoid type offers something specific. Water-rich worlds are the obvious first pick — they turn a colony's scarcest raw material into its surplus. Oil worlds open fuel and plastic at scale. Metal- and volcano-heavy worlds concentrate the ores the asteroid belt scatters. Radioactive worlds are the gate to the endgame: uranium is what the radiation economy runs on, and the richest deposits live on the worlds that emit the most. There is no single best destination — the best destination is the one that produces what your network currently lacks, which is why you will visit several.

A satellite colony is a colony, not a campsite. Landing and claiming a world means giving it oxygen, food, and power of its own — the same three pillars Wave 0 built for the starting pod, now delivered in a rocket and assembled on alien ground. The trick is that you are not there to babysit it: the supply line between worlds is the weakest part of the network, so the goal is an outpost that runs itself from cycle one. That means packing a complete life-support starter — the diffusers or electrolyzers, the farm, the generator and battery — in the first cargo bays, and using the crew's first days to make those systems self-sustaining before any single dupe is allowed to depend on them.

The colony-without-dupes-trips problem is the trap every new world sets. If a planetoid needs a dupe flown in every few cycles to fix a broken pump, the dupe stops being a colonist and becomes a taxi driver, and the network slowly dissolves into a ferry service. Break the cycle at the design stage: an outpost that can lose all its dupes and keep producing — oxygen still made, food still grown, power still generated, doors still closed — is an outpost you can leave. Automation from Wave 1 is the lever: a smart battery pair, a hydro-gated pump, a cooling loop — the same systems that made the base run itself are the systems that let the outpost survive without you.

Warning: Do not settle a world before you can land on it twice. Colonization goes wrong in the first cycles far more often than later — bring a spare rocket's worth of essentials, a backup pilot, and a landing pad at both ends before you commit dupes.

Radiation & Radbolts

Radiation is the second expansion's new physics, layered on top of heat and germs. In Spaced Out, radiation comes from the void above and from radioactive matter below: open space bathes the surface in cosmic radiation, and **uranium ore** and the radioactive world's soil emit steadily. Radiation has two careers in the colony — it harms dupes, and it is a resource.

The resource side is radbolts. A **Radbolt Generator** is a building that, fed radioactive material and power, builds up a charge of **radbolts** — bolts of radiation the machine can hold and then fire. Radbolt beams travel in straight lines and are aimed at receivers: **radbolt reflectors** that turn the beam and, crucially, the **radbolt chamber** and research buildings that accept the charge. This is how the game's highest-tier research is done: radbolt research consumes the bolts in research chambers, and the endgame material-science machines consume them too. The system is deliberately fiddly — a beam blocked by a wall is a beam wasted, so you are building an optical path as much as a machine.

The harm side is radiation sickness. Dupes accumulate **rads** the longer they are exposed — from space, from radioactive ore, from working around radbolt machines. Enough accumulated rads and the dupe develops radiation sickness, a stack of penalties that grows with exposure and eventually becomes genuinely dangerous. The defenses are the ones you expect: **radiation suits** — the Spaced Out counterpart to the exosuit — for dupes who must work in hot zones, shielding for rooms you want to protect, and simple distance: radiation falls off, and a well-placed wall is often the cheapest suit of armor. The numbers — how many rads, at what range, from which source — are the game's tuning, not the lesson. The lesson is that radiation is a territory to be managed, with its own hazmat etiquette.

Warning: Radbolt beams are straight-line and pitiless. Design the beam path — generator to reflector to chamber — before you build, because a single wall between generator and target turns a working research loop into a building that fires into its own ceiling.

Bionic Dupes

The **Bionic Booster Pack** adds a second kind of duplicant to the colony: **bionic dupes**, a distinct duplicant type with requirements that are not the same as a biological dupe's. Where a biological dupe needs air, food, sleep and comfort, a bionic dupe is built on different foundations — which means a colony that hosts them provisions differently than one that does not.

How different, exactly, is something the pack itself teaches better than this guide can. The specifics of what a bionic dupe consumes, what they are immune to, and what they cannot do are the pack's own content, and the in-game duplicant selector and database are the authoritative source. This guide keeps that boundary on purpose: nothing here asserts a number or a building name for bionic dupes that is not verified.

What Wave 2 does assume is the strategic shape. Bionic dupes are an **alternative labor and colonist option**: a colony can draw on biological dupes, bionic dupes, or both. For the network this wave is about, the practical consequence is that the calculus of a new planetoid changes — the life-support an outpost must bring is a function of who will live there. A satellite colony staffed with dupes whose needs differ from air and mealwood may need a very different cargo manifest than one staffed with the crew from Waves 0 and 1. Treat the bionic dupe not as a clone with a paint job but as a genuinely different colonist type, and let the game's own descriptions fill in the specifics before you commit a cargo bay to them.

> **Note:** Bionic dupe mechanics are the pack's content, not this guide's. Check the in-game duplicant database for what a bionic dupe needs before you design an outpost around one — the details are worth reading twice, and this guide deliberately does not print them.

The Endgame

Wave 2 ends when the network stops being a project and becomes a state. The milestones stack like the waves that built them. Research is the first: by the end of Wave 2 the tech tree is meant to be finished, and the last research in Spaced Out is radbolt research — the material-science machines and the final destinations sit behind it. The map is the second: every planetoid worth having is claimed, tamed, and connected, each outpost self-sufficient enough to keep producing while you are not looking. The network is the third: cargo flows on schedule, launchers and rockets run on automation, and no single world's failure can take down the rest.

The explicit goal at the far end is the **temporal tear** — the anomaly in space that Spaced Out's endgame is built around. Traversing it is the game's final objective: it takes the endgame research, the material-science machines, and a mission designed for one last long-haul flight. It is less a fight and more a completion — the whole wave has been building the capability to reach it, and the tear is where that capability is spent.

And then there is the quiet finish: the colony that can keep running without you. Every wave moved a step closer to self-sufficiency — Wave 0's pod that breathed on its own, Wave 1's base that tamed and cooled itself, and now Wave 2's network that flies, ships, and feeds itself on schedule. The true endgame test is to walk away: pause the input, stop nudging dupes, and watch a multiplanetary colony run an entire cycle without a single command. When that holds, the Voyager has finished — not because there is nothing left to build, but because nothing left requires you.

> **Tip:** Treat "finished" as a test, not a moment. Pick one cycle, stop giving orders, and audit what breaks — the outpost that ran dry, the rocket that sat un-fueled, the launcher that clogged. Each break is the last construction project, and the wave is done when nothing breaks.

Wave 2: Modlist

Wave 2 is the capstone, and the mods here are allowed to be ambitious — they exist to deepen the **Spaced Out** experience this wave is built on: rocketry aids that make building and flying rockets less fiddly, planetoid navigation tools that make the star map legible, radiation tooling that makes radbolt work readable, endgame content that stretches the run between the last tech and "done", and performance mods that keep a multi-world colony from becoming a slideshow. Where Wave 1 admitted mods that changed how the game plays, Wave 2 admits mods that change how the **network** plays.

The DLC rule is absolute here: every card must match the DLCs you own — **Spaced Out!** and the **Bionic Booster Pack** — and every card must state its DLC dependency out loud. Mods that require the **Aquatic**, **Frosty** or **Prehistoric** packs are out of scope; you do not own them, and this guide will not include mods you cannot run.

As of this release, every card below is **pending addition**: per project decision, this project does not research or select mods — you do. (The Wave 2 anchor research, recorded in STATUS.md, found no overhaul-scale mod for ONI, so the capstone set is yours to curate.) When you are ready, paste one

verified mod card per mod, with its Workshop URL, its DLC dependency, and a note recording its power-spike verdict.
The mods you add split across three sections by what they change: **Mechanics** for rocket, planetoid, and radiation systems; **Content** for endgame buildings, critters, and recipes; and **Graphics** for overlay and UI clarity across many worlds.

Wave 2: Mechanics

Mechanics mods change how Wave 2's **systems** behave — the rockets you build and fly, the planetoids you navigate between, the radiation you generate and direct. These are STAT-CHANGING mods in the capstone wave, so every one must clear the power-spike gate: no free resources, no free heat deletion, no skipped work. A Wave 2 mechanics mod should make rocketry, colonization, or radiation deeper or clearer — never cheaper.

Wave 2: Content

Content mods add things to build, ranch, grow, or research at the endgame scale — new buildings, new critters, new recipes for the last stretch of Spaced Out. The standing rule is **proportionate cost**: anything powerful must cost what it is worth in materials, power, or labor, or it fails the power-spike gate.

Wave 2: Graphics

Graphics mods change only how the game looks and how it tells you what is happening — the overlays and map clarity you depend on while running a network of worlds. They never touch simulation values, so they carry no power-spike risk; they must simply stay honest — clearer, not smarter.

Glossary

Glossary

Terms populated in Task 13.